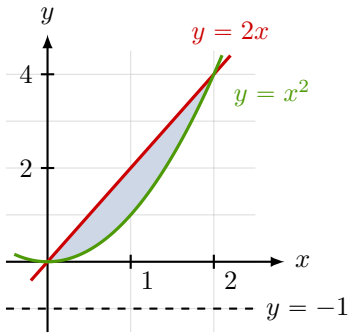
L4.3a Cylindrical Shells

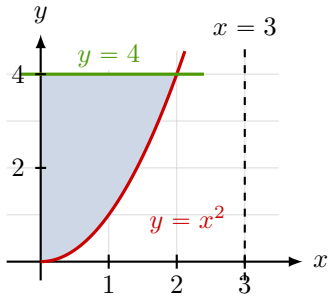
Recall: washers about other lines

Name _____ **Solo – 9 minutes**

1. The region between $y = 2x$ and $y = x^2$ is rotated about the line $y = -1$. Label R and r on the figure and write $A(x)$.



2. The region bounded by $y = x^2$, $y = 4$ and the y-axis is rotated about the line $x = 3$. Write the volume as an integral in y. Do not evaluate.



3. The region under $y = 2x^2 - x^3$, $0 \leqslant x \leqslant 2$, is rotated about the y-axis. A washer is perpendicular to the y-axis, so its radii are written in y. What equation would you have to solve to find them?